

A High Performance Ascon Architecture for Data Encryption and Authentication

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Abstract

The rapid expansion of the Internet of Things (IoT) has led to a massive increase in the number of interconnected embedded devices, making lightweight and efficient cryptographic primitives essential for ensuring data security. Ascon, a standardized lightweight authenticated encryption algorithm, has emerged as a key solution for resource-constrained environments. However, existing hardware implementations of Ascon often suffer from performance degradation and reduced throughput-to-area (TP/A) efficiency, particularly when processing short messages. To address this issue, this paper proposes a novel high-performance hardware architecture for Ascon-128. The proposed design adopts a parallel pipelined ping-pong structure, in which the core permutation module is implemented using a round-based scheme. Through an optimized parallel pipeline control mechanism, the proposed architecture achieves consistent throughput for both short (16 B) and long (1536 B) messages. Experimental results on the Xilinx Kintex-7 FPGA platform demonstrate that the proposed design operates at a maximum frequency of 187 MHz and achieves a peak throughput of 8988 Mbps. Compared with state-of-the-art pipelined implementations, the proposed architecture improves the operating frequency by 78%, the throughput by 33.7%, and the throughput-to-area ratio by 41.6%.

CCS Concepts

• **Hardware** → Integrated circuits; Logic circuits.

Keywords

Ascon, FPGA, Pipeline, Encryption

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1 Introduction

With the rapid growth of embedded devices and sensors connected to the Internet of Things (IoT), a large volume of sensitive data is continuously transmitted across networks. These devices often store and process private and mission-critical information, as exemplified by the Internet of Medical Things (IoMT) and smart manufacturing systems. For example, wearable medical devices collect physiological data from patients [1], while smart factories continuously monitor equipment status, generating vast amounts of critical data. Consequently, ensuring secure data transmission is essential to mitigate the associated security and privacy risks [2].

Most conventional cryptographic standards have been in use for over two decades and are no longer well suited to the emerging requirements of the Internet of Things and edge devices. To address this challenge, the National Institute of Standards and Technology (NIST) initiated the Lightweight Cryptography (LWC) project for constrained environments [3], completed its evaluation process in 2023, and selected Ascon as the standardized LWC algorithm [4]. The Ascon cipher suite [5] supports authenticated encryption with associated data (AEAD), including ASCON-128 and ASCON-128a. All variants provide 128-bit security and share the same 320-bit permutation core, differing only in the number of permutation rounds.

Various hardware implementation strategies for Ascon have been reported, including unrolled, round-based, and serialized architectures [6]. The work in [7] proposes a recursive permutation structure that enables two permutation rounds to be executed within a single clock cycle. Elsadek *et al.* [8] evaluated the throughput performance of AEAD implementations under six different operating scenarios. In [9], a round-based implementation was proposed, in which a single hardware module supports both encryption and decryption operations. The work in [10] achieved an improved throughput-to-area ratio by increasing the number of permutation rounds per cycle. The hardware architecture presented in [11] adopts a phased full-pipeline design.

In this work, a novel high-performance hardware implementation architecture for Ascon-128 is proposed. The proposed design adopts a parallel pipelined architecture that achieves consistent throughput for both short (16 B) and long (1536 B) messages. Furthermore, the architecture follows a round-based implementation strategy, in which the same hardware module supports both encryption and decryption modes. While achieving high throughput, the proposed architecture also maintains a relatively high throughput-to-area ratio.

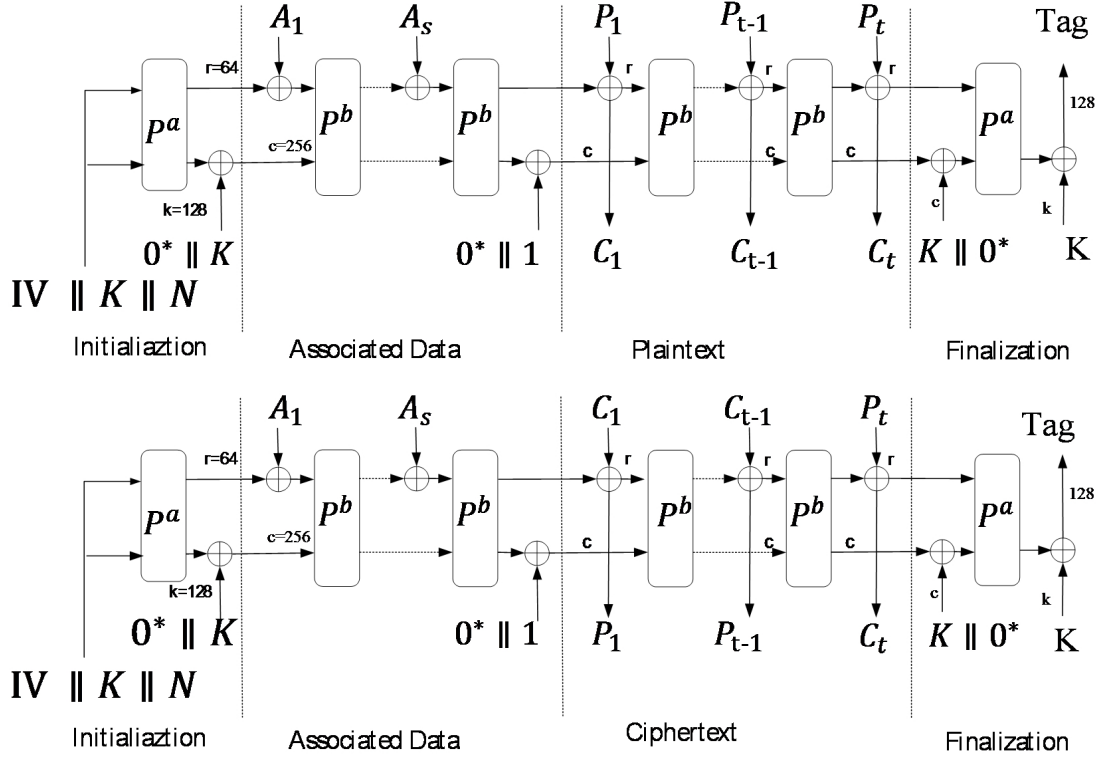


Figure 1: Encryption and Decryption architecture

Table 1: Parameters of Ascon-128 and Ascon-128a.

Parameter	Ascon-128	Ascon-128a
IV length (bits)	64	64
Key length (bits)	128	128
Nonce length (bits)	128	128
Rate r (bits)	64	128
Capacity c (bits)	256	192
Tag length (bits)	128	128
Permutation rounds (p^a)	12	12
Permutation rounds (p^b)	6	8

2 ascon fundamentals

Ascon is a lightweight symmetric-key cryptographic algorithm that supports authenticated encryption with associated data (AEAD) [4]. The CAESAR competition ultimately designated Ascon-128 and Ascon-128a from the Ascon cipher suite as recommended choices for lightweight authenticated encryption. The encryption and decryption operations of the Ascon AEAD algorithm are defined in eq (1) and (2), respectively [4].

$$\mathcal{E}_{k,r,a,b}(K, N, A, P) = (C, T). \quad (1)$$

$$\mathcal{D}_{k,r,a,b}(K, N, A, C, T) \in \{P, \perp\}. \quad (2)$$

where K (key, ≤ 160 bits), N (nonce, 128 bits), and T (tag, 128 bits) have fixed lengths, whereas A (associated data) and P (plaintext)

are of variable length. The ciphertext C has the same length as the plaintext P . If tag verification fails, the decryption process outputs $\perp$.

Table 1 summarizes the differences between these two variants, including the data rate, the number of permutation rounds, and the tag bit length of each variant [8]. Figure 1 illustrates the encryption and decryption processes of Ascon, which consist of four stages: initialization, associated data processing, plaintext–ciphertext conversion, and finalization.

2.1 Ascon Permutation

The permutation function $p(\cdot)$ constitutes the core component of the Ascon cipher. As shown in Figure 1, all stages of Ascon algorithm need replacement operations, including internal state S is divided into five 64-bit word x_i .

$$S = x_0 \parallel x_1 \parallel x_2 \parallel x_3 \parallel x_4$$

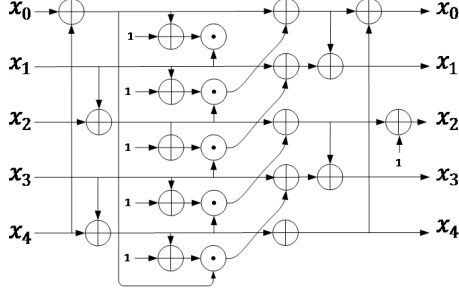
Addition of constants: During each permutation round, the corresponding round constant c_r listed in Table 2 is XORed with the state word x_2 .

$$x_2 = x_2 \oplus c_r$$

Substitution layer: This layer is the only nonlinear component of the algorithm and is designed to provide resistance against differential and linear attacks. Ascon-128 employs a 5×5 substitution box (S-Box), whose inputs are derived from the five state words

Table 2: Round Constants c_r used in the Ascon permutation

p^{12}	p^8	p^6	Constant c_r	p^{12}	p^8	p^6	Constant c_r
0	0	0	$0^* \parallel f0$	6	2	0	$0^* \parallel 96$
1	0	0	$0^* \parallel e1$	7	3	1	$0^* \parallel 87$
2	0	0	$0^* \parallel d2$	8	4	2	$0^* \parallel 78$
3	0	0	$0^* \parallel c3$	9	5	3	$0^* \parallel 69$
4	0	0	$0^* \parallel b4$	10	6	4	$0^* \parallel 5a$
5	1	0	$0^* \parallel a5$	11	7	5	$0^* \parallel 4b$

**Figure 2: Ascon's 5-bit S-box**

x_0, x_1, x_2, x_3 and x_4 . The corresponding S-box circuit is illustrated in Figure 2.

Linear diffusion layer: The permutation round is completed by applying the following shift and XOR operations to each state word x_i .

$$\begin{aligned}
 x_0 &\leftarrow x_0 \oplus (x_0 \gg 19) \oplus (x_0 \gg 28) \\
 x_1 &\leftarrow x_1 \oplus (x_1 \gg 61) \oplus (x_1 \gg 39) \\
 x_2 &\leftarrow x_2 \oplus (x_2 \gg 1) \oplus (x_2 \gg 6) \\
 x_3 &\leftarrow x_3 \oplus (x_3 \gg 10) \oplus (x_3 \gg 17) \\
 x_4 &\leftarrow x_4 \oplus (x_4 \gg 7) \oplus (x_4 \gg 41)
 \end{aligned}$$

2.2 Initialization

The initial state consists of a 64-bit initialization vector (IV), a 128-bit key K, and a 128-bit nonce N. The IV values and the construction of the initial state S for different Ascon variants are defined as follows.

$$\text{Ascon} - 128, \quad IV_{k,r,a,b} = 0x80400c6000000000.$$

$$\text{Ascon} - 128 - a, \quad IV_{k,r,a,b} = 0x80400c8000000000$$

$$S = IV \parallel K \parallel N$$

Subsequently, the same number of full-round permutation functions p^a are applied to both Ascon-128 and Ascon-128a variants, namely 12 rounds. After the permutation operations, the 128-bit key K is XORed with the least significant 128 bits of the internal state. The state update can be expressed as:

$$S \leftarrow p^a(S) \oplus (0^* \parallel K)$$

2.3 Process Associated Data

In Ascon-128, the data block size is 64 bits, and each block is XORed with the internal state. Each associated data block A_i is XORed with the most significant 64-bit of the internal state, followed by a 6-round permutation operation. This process is repeated until all associated data blocks have been processed.

$$S \leftarrow p^b((S_r \oplus A_i) \parallel S_c), \quad 1 \leq i \leq s$$

Finally, the internal state is XORed with a single-bit value 1.

$$S \leftarrow S \oplus (0^{319} \parallel 1)$$

2.4 Processing of Plaintext and Ciphertext

This stage operates in a manner similar to the preceding stage; however, the key difference is that the ciphertext or plaintext is generated as part of the XOR operation.

2.4.1 Process Plaintext. In each iteration, a plaintext block P_i is XORed with the rate part of the internal state to generate the corresponding ciphertext block, followed by a p^b round of permutation operation except for the final block. For Ascon-128, the rate is 64 bits. The final plaintext block is processed without applying an additional permutation operation.

$$C_i \leftarrow S_r \oplus P_i$$

$$S = \begin{cases} p^b(C_i \oplus S_c), & \text{if } 1 \leq i < t \\ C_i \oplus S_c, & \text{if } 1 \leq i = t \end{cases}$$

2.4.2 Process Ciphertext. The decryption process follows a procedure analogous to encryption and exploits the computational properties of the XOR operation for efficient implementation. For each ciphertext block C_i , the plaintext block P_i is obtained by XORing C_i with the rate part of the internal state. Subsequently, a p^b round permutation is applied update the internal state, concluding the iteration. After all iterations are completed, the full plaintext is recovered.

$$P_i \leftarrow S_r \oplus C_i$$

$$S \leftarrow p^b(C_i \parallel S_c)$$

2.5 Finalization

In the finalization stage, the internal state S is XORed with the key K, followed by a p^a round permutation operation. The authentication tag T is generated by XORing the least significant 128 bits of the internal state with the key K.

$$S \leftarrow p^a(S \oplus (0^r \parallel K \parallel 0^{c-k}))$$

$$T \leftarrow [S]^{128} \oplus [K]^{128}$$

3 Hardware Architecture of Ascon

In this section, a detailed description of the proposed hardware architecture is presented. The permutation module serves as the core component of the Ascon algorithm and is required in all processing stages. As presented in [6], a round-based implementation of Ascon-128 updates the internal state by executing one permutation per clock cycle. In round-based architectures, a one-round-per-cycle design achieves the highest throughput-to-area (TP/A) ratio, whereas multi-round-per-cycle designs provide higher peak throughput. To

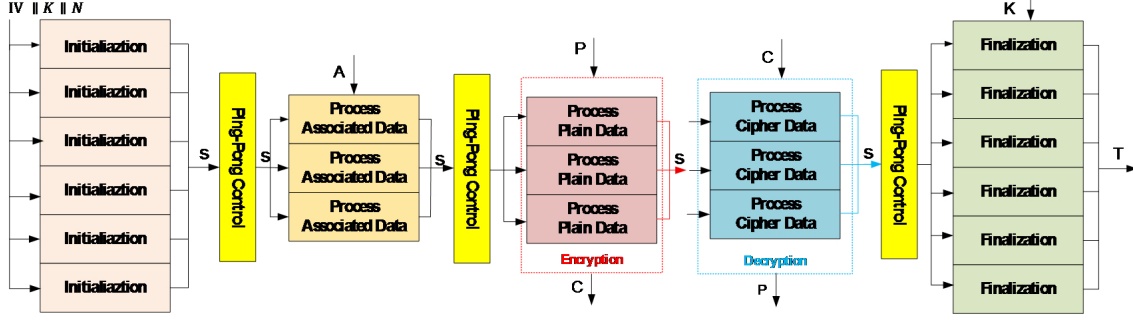


Figure 3: Parallel Ping-Pong Architecture of Ascon-128

improve authenticated encryption performance while maintaining efficient resource utilization, the architecture shown in Figure 3 adopts a parallel ping-pong design, in which all stages of the Ascon algorithm employ a single permutation round per clock cycle.

For Ascon-128, the initialization and finalization phases employ the p^a permutation round, requiring 14 clock cycles to propagate the updated internal state to the subsequent stage. Meanwhile, the associated data processing and encryption/decryption (Enc/Dec) phases utilize the p^b permutation, which requires 8 clock cycles to update the internal state. Designed to output three 128-bit ciphertext blocks every 8 clock cycles, the architecture shown in Figure 3 employs six initialization/finalization modules, three associated data and plaintext/ciphertext processing modules, and three ping-pong control modules. This design achieves uniform throughput for messages ranging from short (16-byte) to long (1536-byte) lengths, thereby overcoming the short-message efficiency degradation commonly observed in existing Ascon hardware implementations.

4 RESULTS AND DISCUSSION

This section presents the implementation results of the proposed Ascon-128 architecture and compares them with those of other pipelined designs. Each implementation is evaluated in terms of LUT utilization, operating frequency, throughput, and throughput-to-area (TP/A) ratio. Throughput (TP), defined as the number of encrypted bits per second, is calculated using Eq 3.

$$\text{Throughput} = \frac{\text{No. of bits processed}}{\text{period} \times \text{Clock Cycle}} \quad (3)$$

Since different implementations may use different associated data and plaintext sizes, throughput and throughput-to-area (TP/A) ratio are important evaluation metrics. To verify functional correctness, a co-simulation environment was established using the official Ascon C implementation as the golden reference. A custom test-bench automatically compares the Verilog design's outputs against the C model. Testing utilized the official Ascon Known Answer Test suite (for Ascon-128) covering operations with empty data, boundary lengths, and varied inputs, supplemented by extensive random vectors. This confirms full functional compliance with the standard.

The implementation results of the proposed architecture and those reported in [11] are summarized in Table 3. Both implementations were synthesized and evaluated on a Xilinx Kintex FPGA using

Vivado. Compared with the implementation in [11], the proposed design achieves a 78% increase in operating frequency, along with improvements of 33.7% in throughput and 41.6% in throughput-to-area ratio. One reason for the comparable LUT utilization between the two designs is that both implementations employ a similar number of basic round-permutation units. The key architectural difference is that the design in [11] executes three or even four permutation rounds per clock cycle, whereas the proposed architecture operates multiple permutation units in parallel, each performing a single round per cycle. As a result, although a more complex ping-pong control logic is introduced, the proposed design achieves a significant performance improvement, particularly in operating frequency.

A prevailing characteristic of IoT communication is the frequent, periodic exchange of small data packets. For example, a typical data packet—containing device ID, sensor type, data value, and timestamp—usually has a size of around 16 bytes. The proposed design achieves the highest throughput among all compared implementations for message lengths ranging from 8 bytes to 1536 bytes. Although the design reported in [10] exhibits the highest TP/A ratio, it reaches its peak throughput only when processing long messages (e.g., 1536 bytes). Implementations based on finite-state machine architectures, such as [8], suffer a significant throughput degradation—up to 56%—when processing short messages.

5 Conclusion

The National Institute of Standards and Technology (NIST) has selected Ascon as a standardized lightweight Authenticated Encryption with Associated Data (AEAD) scheme. Consequently, a high-performance hardware architecture capable of efficiently processing messages of varying lengths is required. This work proposes a high-performance hardware implementation of Ascon that focuses on optimizing plaintext processing efficiency while achieving a high throughput-to-area (TP/A) ratio, and achieves a peak throughput of 8988 Mbps. Compared with the evaluated designs in the literature, the proposed implementation demonstrates improved throughput and throughput-to-area performance. Moreover, the architecture maintains stable throughput across different plaintext lengths and provides enhanced flexibility for diverse encryption and authentication scenarios. The proposed design achieves high processing efficiency and demonstrates strong adaptability to

Table 3: Comparison of FPGA-based Ascon AEAD implementations

Design	Device	Freq(MHz)	LUT	TP(Mbps)	TP/A ratio(Mbps/LUT)
IOTJ [6]	Spartan-6	112.3	4971	899.17	0.183
TRON’22 [7]	Virtex-4	189	4021	506.29	0.125
ISCAS’25 [8]	Spartan-7	—	805	1180	1.46
VLSID’24 [9]	Virtex-7	173.7	1723	729	0.423
ISCT’24 [10]	Kintex-7	92.59	2958	5925.92	2.01
VLSI [11]	Kintex-7	105	14200	6720	0.473
This work	Kintex-7	187	13472	8988	0.670

variable-length data streams, enabling high-speed and real-time secure data transmission in a wide range of IoT applications.

Acknowledgments

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References

- [1] K. Raj and S. Bodapati, "FPGA Based Light Weight Encryption of Medical Data for IoMT Devices using ASCON Cipher," 2022 IEEE International Symposium on Smart Electronic Systems (iSES), Warangal, India, 2022, pp. 196-201, doi: 10.1109/iSES54909.2022.00048.
- [2] W. Zhou, Y. Jia, A. Peng, Y. Zhang and P. Liu, "The Effect of IoT New Features on Security and Privacy: New Threats, Existing Solutions, and Challenges Yet to Be Solved," in IEEE Internet of Things Journal, vol. 6, no. 2, pp. 1606-1616, April 2019, doi: 10.1109/JIOT.2018.2847733.
- [3] NIST, Lightweight Cryptography: Project Overview, <https://csrc.nist.gov/projects/lightweight-cryptography>. 2025.
- [4] C. Dobraunig, M. Eichlseder, F. Mendel, and M. Schl  ffer, "Ascon v1.2: Lightweight authenticated encryption and hashing," Journal of Cryptology, vol. 34, no. 3, 2021. doi:10.1007/s00145-021-09398-9.
- [5] M. S. Turan, *et al.*, "Status Report on the Final Round of the NIST Lightweight Cryptography Standardization Process," NIST Internal Report, NIST IR 8454, June 2023.
- [6] S. Khan, W. -K. Lee and S. O. Hwang, "Scalable and Efficient Hardware Architectures for Authenticated Encryption in IoT Applications," in IEEE Internet of Things Journal, vol. 8, no. 14, pp. 11260-11275, 15 July, 2021, doi: 10.1109/JIOT.2021.3052184.
- [7] S. Khan, W. -K. Lee and S. O. Hwang, "Evaluating the Performance of Ascon Lightweight Authenticated Encryption for AI-Enabled IoT Devices," 2022 TRON Symposium (TRONSHOW), Tokyo, Japan, 2022, pp. 1-6.
- [8] I. Elsadek and E. Tawfik, "Optimized and Reconfigurable Hardware Design for ASCON AEAD and Hash Standards," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-5, doi: 10.1109/ISCAS56072.2025.11043658.
- [9] A. Koppuravuri, H. Pasupuleti, S. Gvk and J. Bapat, "A High Throughput ASCON Architecture for Secure Edge IoT Devices," 2024 37th International Conference on VLSI Design and 2024 23rd International Conference on Embedded Systems (VLSID), Kolkata, India, 2024, pp. 486-491, doi: 10.1109/VLSID60093.2024.00087.
- [10] M. Ahmet, "High-Performance FPGA Implementations of Lightweight ASCON-128 and ASCON-128a With Enhanced Throughput-to-Area Efficiency," 2024 17th International Conference on Information Security and Cryptology (ISCT  rkiye), Ankara, Turkiye, 2024, pp. 1-7, doi: 10.1109/ISCTrkiye64784.2024.10779273.
- [11] D. Xu, X. Wang, Q. Hao, J. Wang, S. Cui and B. Liu, "A High-Performance Transparent Memory Data Encryption and Authentication Scheme Based on Ascon Cipher," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 32, no. 5, pp. 925-937, May 2024, doi: 10.1109/TVLSI.2024.3372026.